CAS Case Competition

PRESENTED BY

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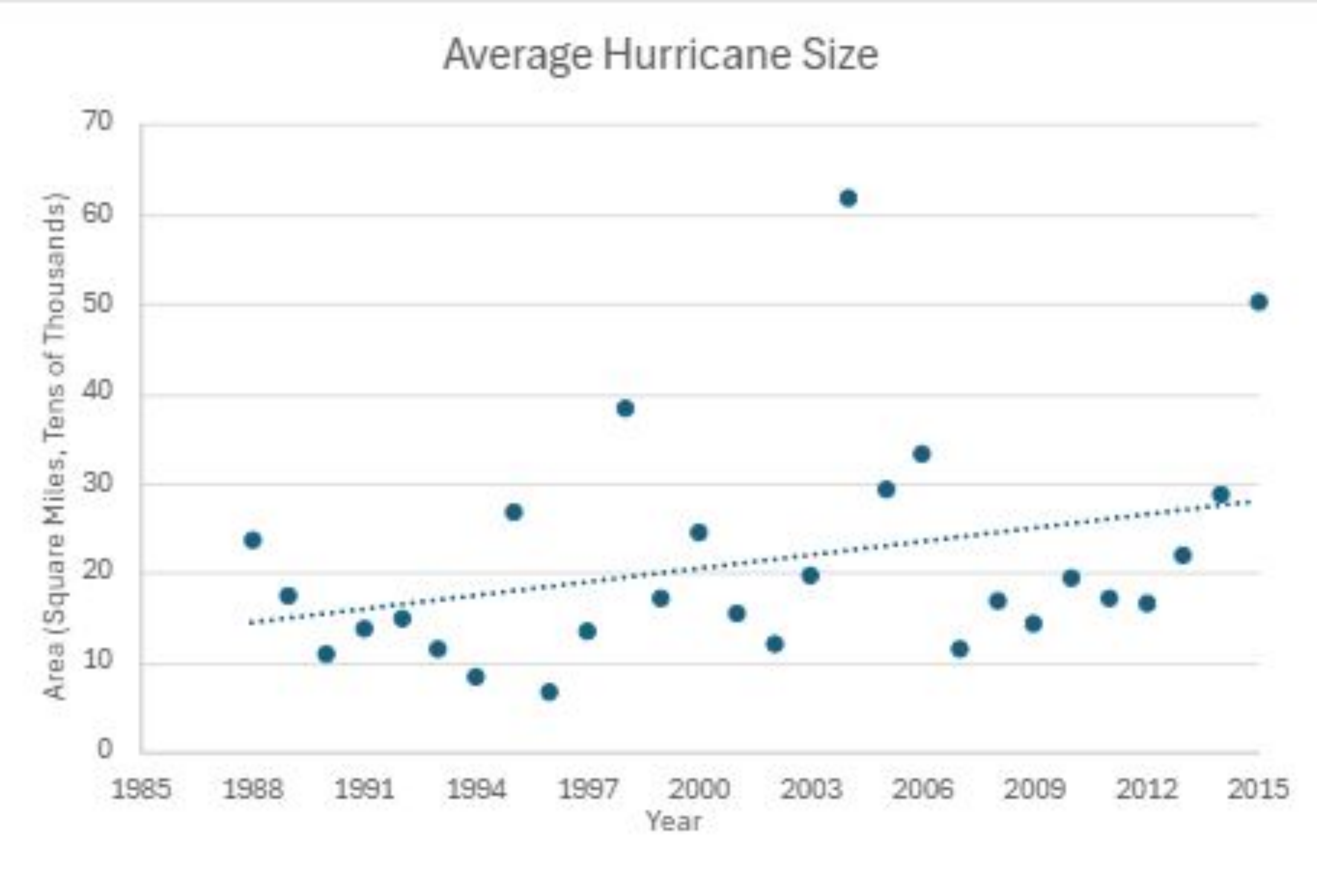
CAS(TASTROPHE)

Climate Change

The increase in severe weather events is likely set to significantly impact the insurance industry by affecting the ability of underwriters to measure, predict and apportion risks.

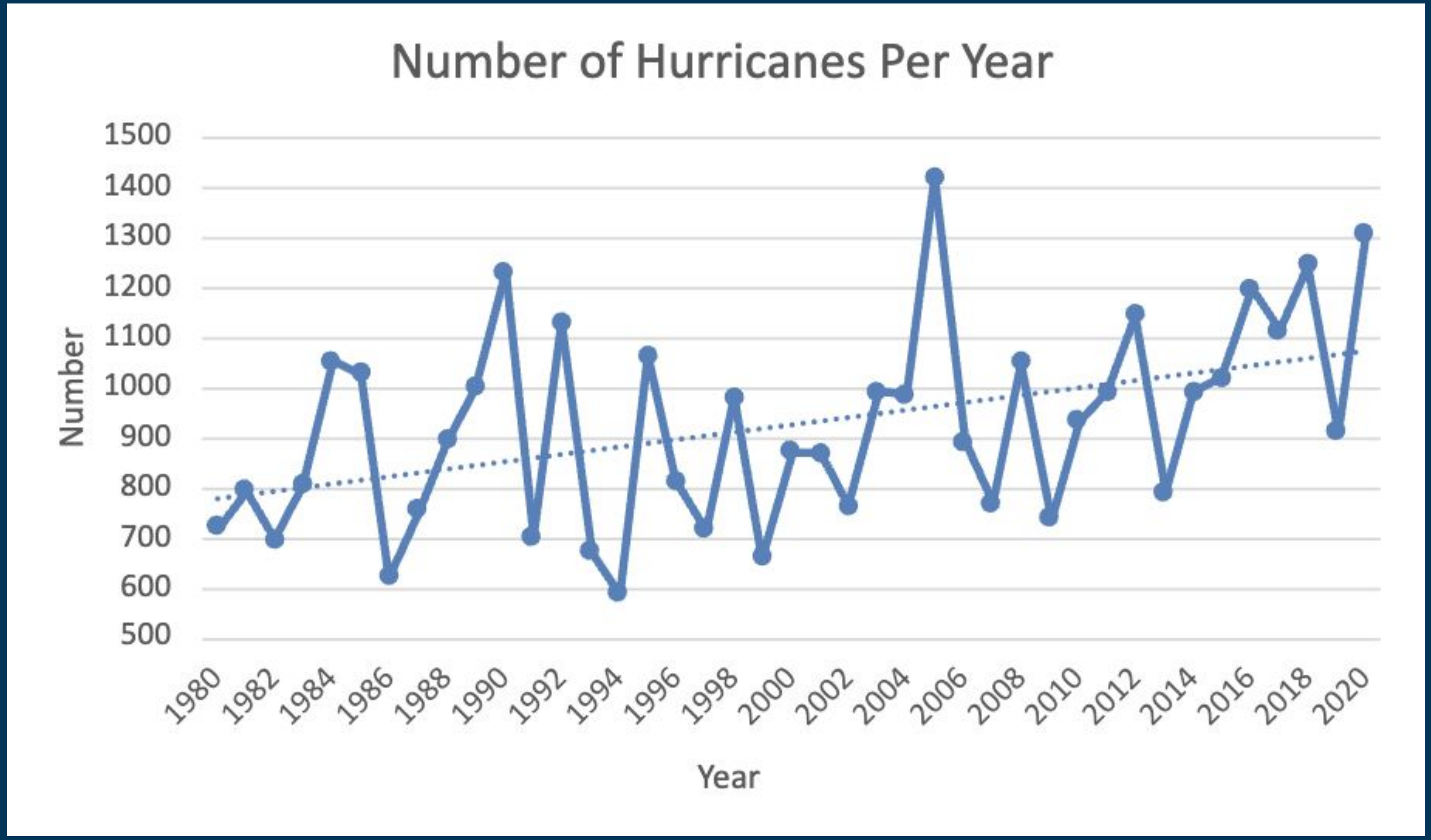
Effects on Property Insurance

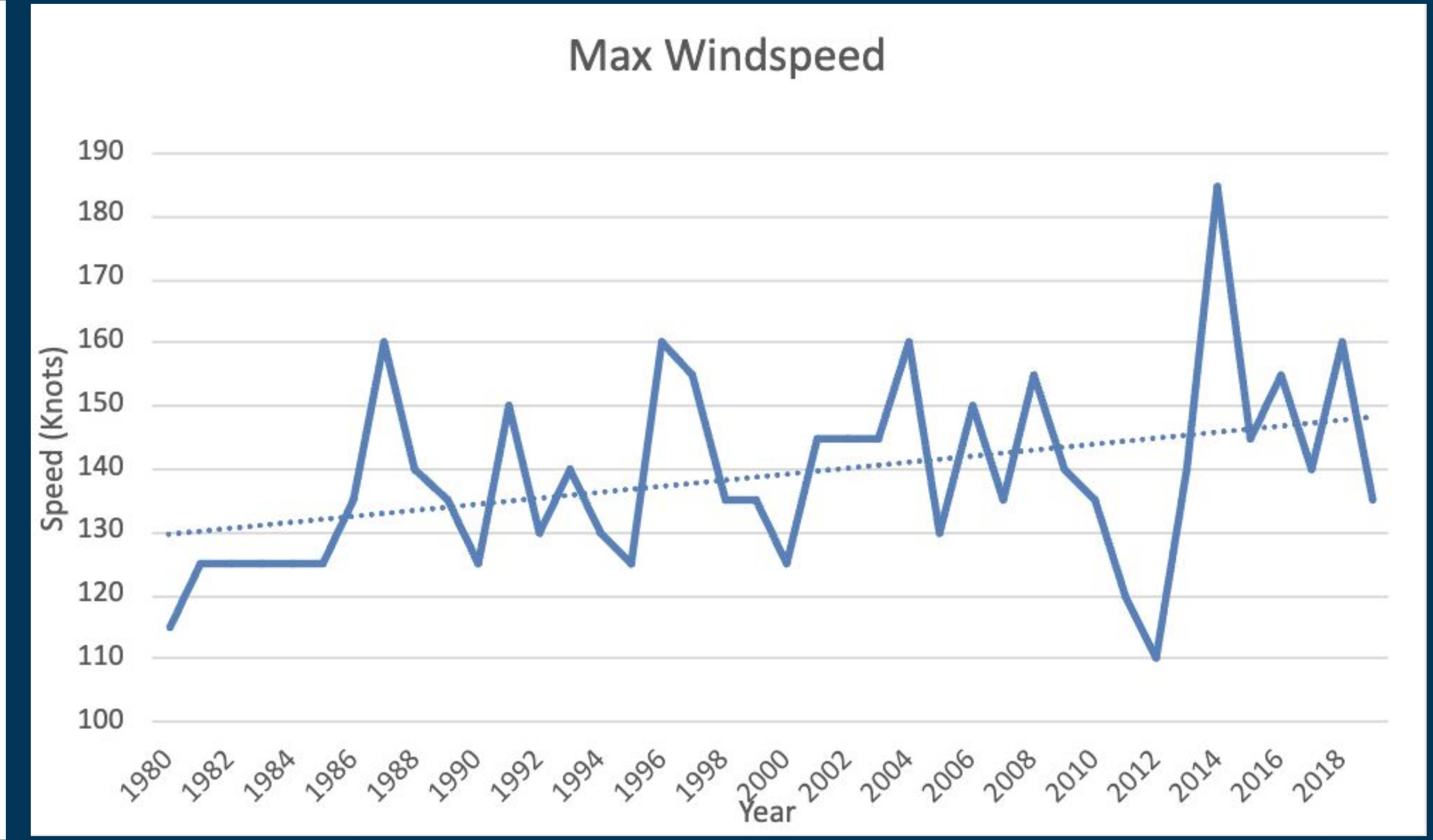
- Rising claim costs for insurers
- Increasing premiums for policyholders
- Increasing frequency and severity of catastrophic events



Hurricane Trends

- These trends result in hurricanes that are more costly in physical damages



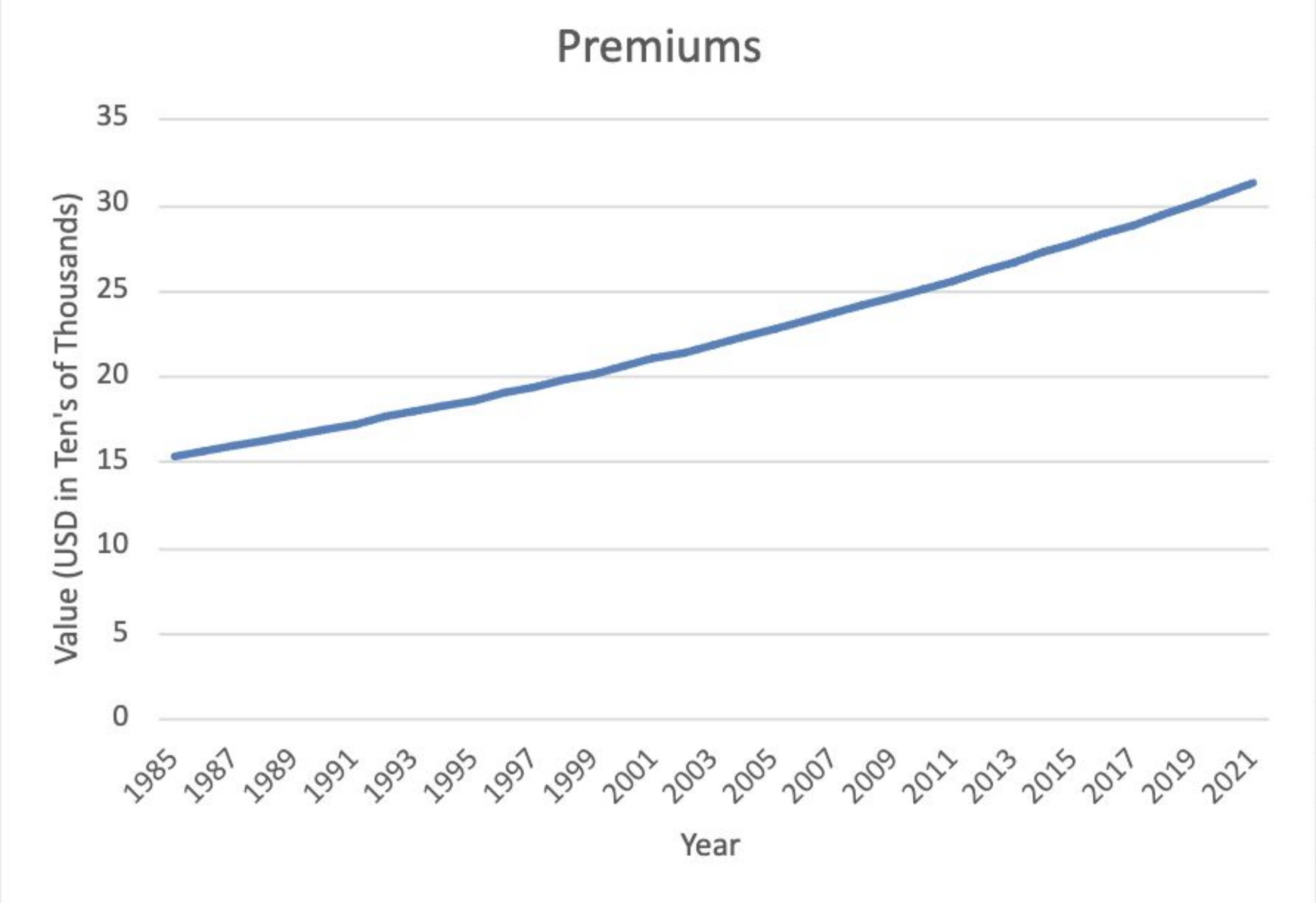


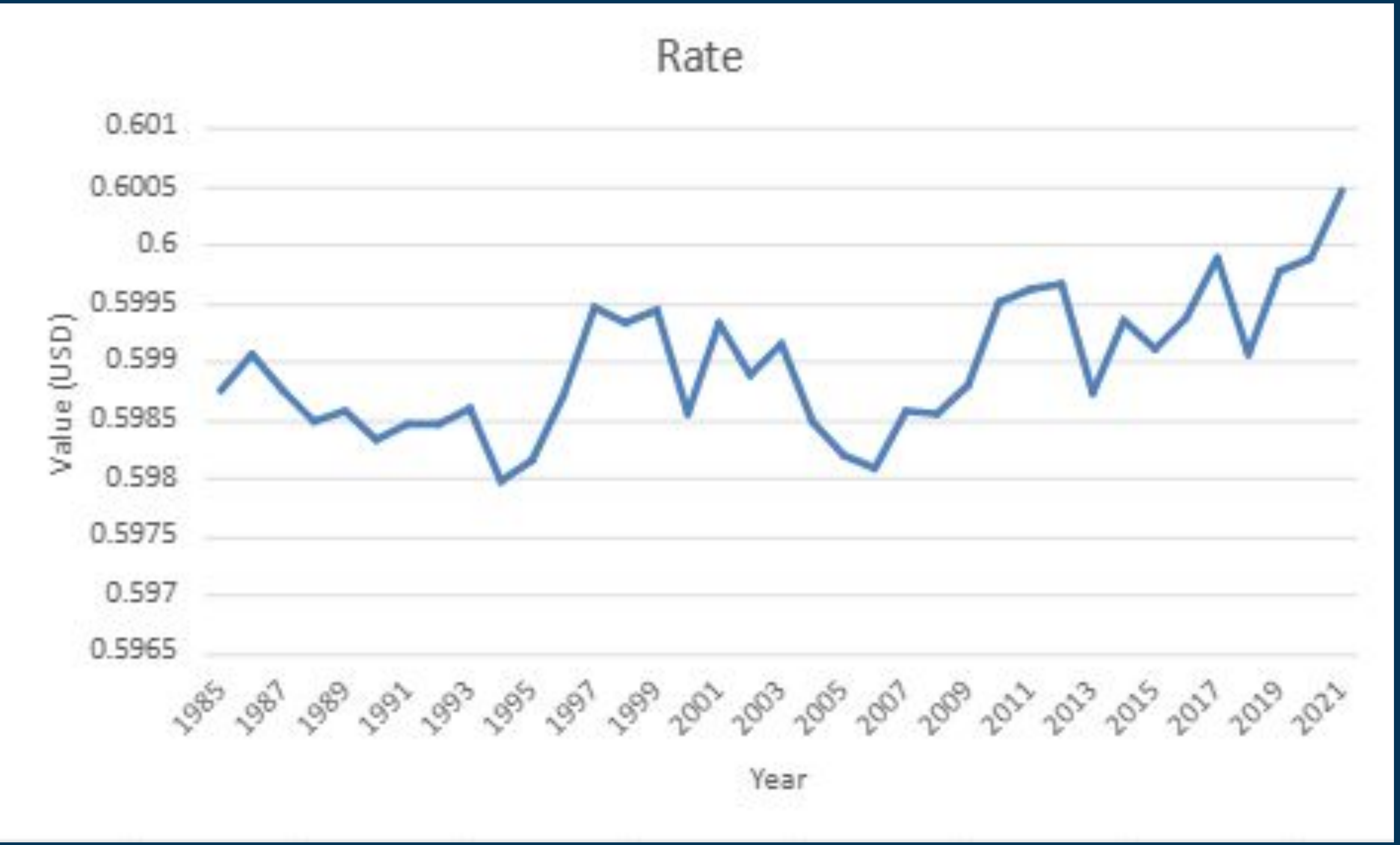
Premiums

- Premiums have been steadily increasing
- Due to increased claim costs, inflation, climate change, etc.

Rate

- The rate shows an increasing trend
- Insurers are charging more for coverage due to increased risk of damages





*Note that "Rate" refers to premium per $100 of total insured value

Loss Ratios (Excluding CAT Losses)

Year	Loss ratios Excl CAT
1985	26%
1986	162%
1987	156%
1988	10%
1989	58%
1990	39%
1991	52%
1992	115%
1993	49%
1994	48%
1995	6%
1996	63%
1997	59%
1998	32%
1999	148%
2000	154%
2001	23%
2002	8%
2003	93%
2004	91%
2005	20%

Year	Loss ratios Excl CAT
2006	48%
2007	135%
2008	35%
2009	14%
2010	65%
2011	94%
2012	46%
2013	35%
2014	45%
2015	26%
2016	68%
2017	154%
2018	17%
2019	76%
2020	29%
2021	107%

- CAT events: Low frequency, high severity
- Excluding CAT for better representation of profitability and performance

Estimating CAT Losses (Assumptions)

Assumptions:

- Only CAT events that occurred are hurricanes
- Adjusted the Saffir-Simpson Scale* to group Hurricanes into 3 categories
- Hurricanes are independent of each other

Category	Wind Speed (Knots)	Estimated Probability of Damage
Low	< 70	0.05
Medium	70 - 112	0.3
High	> 112	0.7

*A commonly used scale used to categorize hurricanes into 5 categories.

Estimating CAT Losses

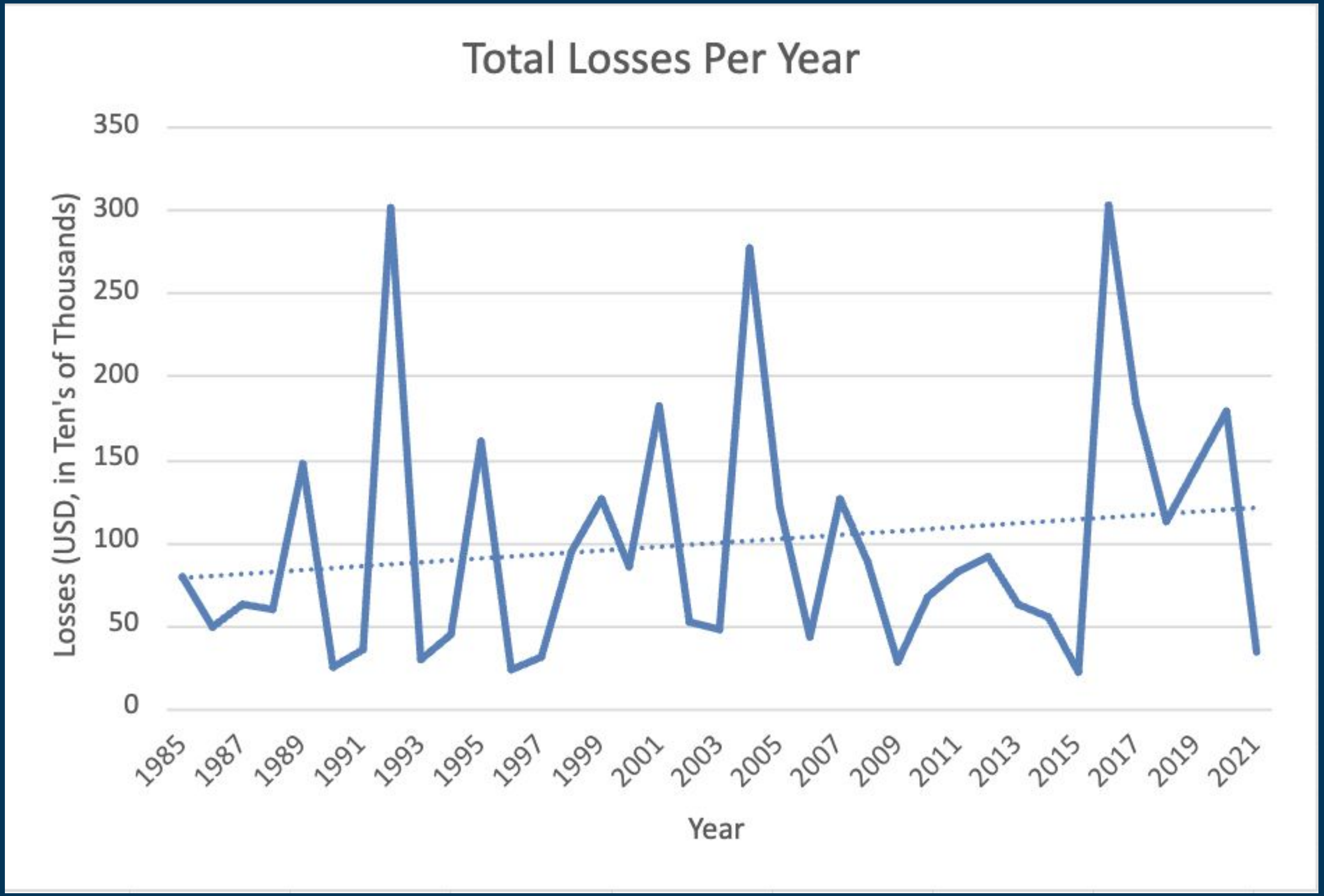
- If location is within 1 degrees longitude and latitude from a historical hurricane, we assume that the property is damaged by the hurricane

- Losses to damaged properties is calculated by multiplying Probability of Damage and Total Insured Value

Location	PolicyYear	Wind Speed	Category	Probability Damaged	Losses - Catastrophe	Total Losses
14	2000	30	Low	0.05	12,256	12,256
14	2001	75	Medium	0.3	74,955	74,955
14	2002	30	Low	0.05	12,729	12,729
14	2003	0	Low	0	-	-
14	2004	25	Low	0.05	13,208	13,208
14	2005	100	Medium	0.3	80,665	80,665
14	2006	0	Low	0	-	-
14	2007	125	High	0.7	196,168	196,168
14	2008	60	Low	0.05	14,253	14,253
14	2009	0	Low	0	-	-
14	2010	0	Low	0	-	-
14	2011	0	Low	0	-	-
14	2012	75	Medium	0.3	92,131	92,131
14	2013	0	Low	0	-	-
14	2014	0	Low	0	-	-

Note: “At risk” is a measure defined by our client insurance company.

Losses



Loss Ratios (Including CAT)

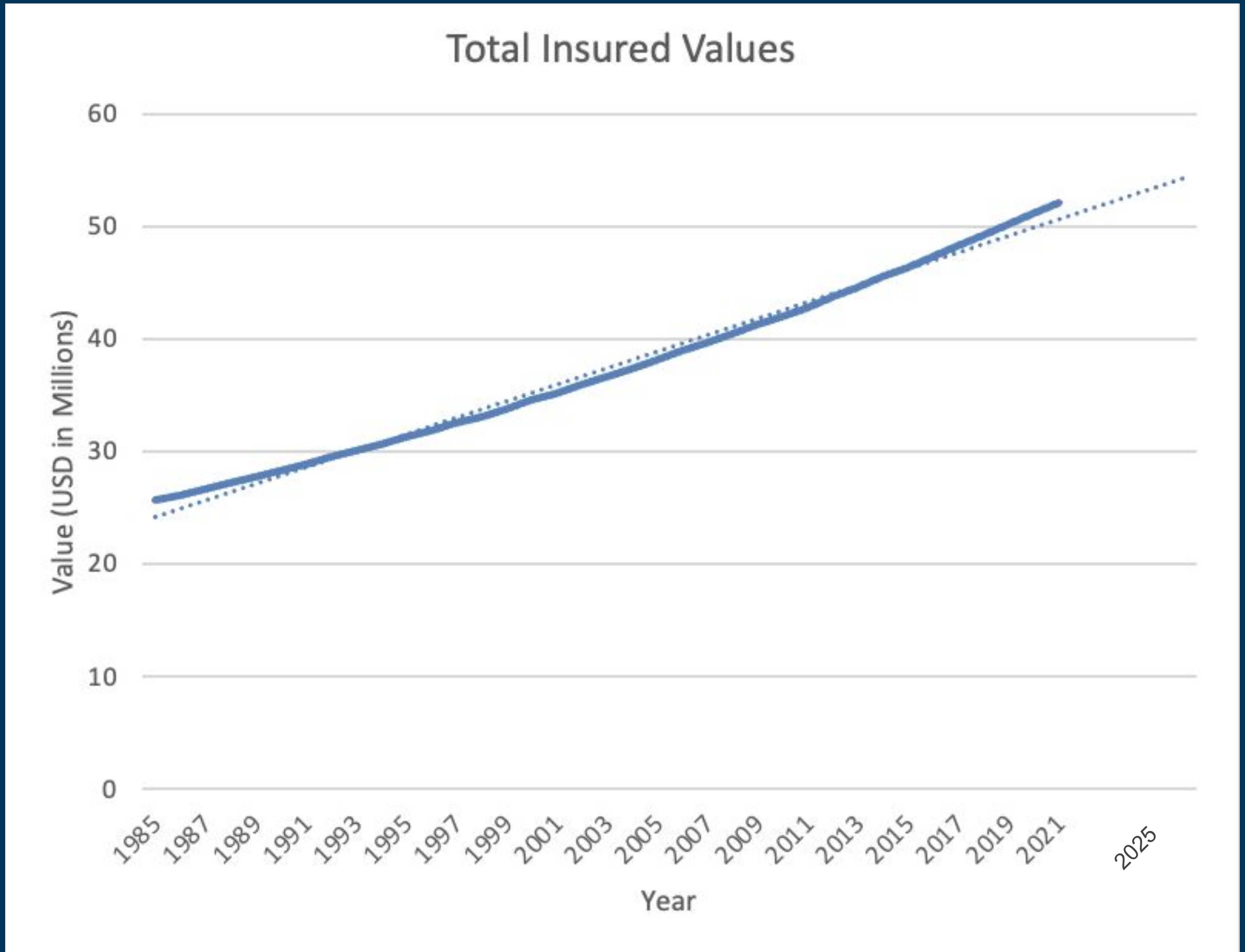
- Fluctuating Loss ratios across the period

Year	Loss Ratio
1985	520%
1986	318%
1987	399%
1988	366%
1989	893%
1990	147%
1991	207%
1992	1718%
1993	164%
1994	247%
1995	861%
1996	123%
1997	161%
1998	479%
1999	627%
2000	412%
2001	864%
2002	243%
2003	216%
2004	1244%

Year	Loss Ratio
2005	537%
2006	189%
2007	533%
2008	366%
2009	117%
2010	269%
2011	319%
2012	347%
2013	235%
2014	204%
2015	78%
2016	1071%
2017	635%
2018	384%
2019	485%
2020	586%
2021	107%

Total Insured Value

- Total insured value has been steadily increasing
- Insurer is writing more policies
- Coverage is becoming more expensive
- Higher exposure to risk
- External factors, eg. inflation, economic development



Historical Exposures to Hurricane by Property Location

Location	Latitude	Longitude	Number of storms
1	19	-100.1	2
2	7.2	-74.4	0
3	24.7	-77.8	16
4	40.5	-122.3	0
5	29.6	-81.3	20
6	29.6	-90.6	16
7	32.6	-83.4	13
8	43.4	-107.9	0
9	51.5	-73.7	0
10	46.2	-62.5	7
11	17.2	-89	13
12	34.6	-77.9	26
13	30.4	-87.5	19
14	18	-77.2	10
15	41.9	-87.9	1
16	26.5	-97.8	11
17	33.8	-80.7	17
18	24.8	-111.6	19
19	51.1	-114.3	0
20	25	-77.4	16
21	35.6	-87	7
22	39.4	-94.8	0
23	4.8	-52.5	0
24	15	-83.7	17
25	35.4	-90.9	6
26	39.7	-86.1	2
27	33	-93.6	12
28	37.9	-88.7	5
29	25.9	-81.7	17
30	19.5	-99.2	5
31	-0.3	-79.3	0
32	-9.3	-71.1	0
33	-15.7	-61.7	0
34	40.7	-73.9	10
35	44.3	-68	4

Legend

: Properties that have been exposed to 15 or more hurricanes since 1985

Current Total Insured Value at Risk (2025)

Location	Total Insured Value	Risk condition
3	$ 2,333,477.84	At risk
5	$ 1,975,268.33	At risk
6	$ 1,082,568.92	At risk
7	$ 2,021,698.87	At risk
10	$ 1,582,108.26	At risk
11	$ 789,128.06	At risk
12	$ 542,205.13	At risk
13	$ 2,021,063.60	At risk
16	$ 587,539.61	At risk
17	$ 3,498,666.70	At risk
20	$ 2,945,901.55	At risk
21	$ 3,248,652.39	At risk
24	$ 415,002.58	At risk
25	$ 4,567,173.40	At risk
27	$ 1,542,377.55	At risk
28	$ 2,099,516.57	At risk
29	$ 681,526.39	At risk
34	$ 959,973.01	At risk

Assumptions:

- "At risk": locations within 1 degree longitude and latitude from the historical hurricane as defined by the insurance company
- Total insured value follows the same growth pattern for each years

Total Insured Value of all properties: ***$53,532,904.04***
Total Insured Value of properties at risk: ***$32,893,848.73***

61.45% of the Total Insured Value is at risk

Expected Losses

Estimate Expected Losses of Properties Currently "At Risk"

Location	PL Category	Total Insured Value	Prob	Expected Loss
3	Medium	$ 2,333,477.84	0.3	$ 700,043.35
5	Low	$ 1,975,268.33	0.05	$ 98,763.42
6	Medium/Low	$ 1,082,568.92		$ 378,899.12
7	Low	$ 2,021,698.87	0.05	$ 101,084.94
10	Low	$ 1,582,108.26	0.05	$ 79,105.41
11	Low	$ 789,128.06	0.05	$ 39,456.40
12	Low	$ 542,205.13	0.05	$ 27,110.26
13	Medium	$ 2,021,063.60	0.3	$ 606,319.08
16	Medium	$ 587,539.61	0.3	$ 176,261.88
17	Low	$ 3,498,666.70	0.05	$ 174,933.34
20	Medium	$ 2,945,901.55	0.3	$ 883,770.47
21	Low	$ 3,248,652.39	0.05	$ 162,432.62
24	Low/High	$ 415,002.58		$ 311,251.93
25	Low	$ 4,567,173.40	0.05	$ 228,358.67
27	Low	$ 1,542,377.55	0.05	$ 77,118.88
28	Low	$ 2,099,516.57	0.05	$ 104,975.83
29	Low	$ 681,526.39	0.05	$ 34,076.32
34	Low	$ 959,973.01	0.05	$ 47,998.65

Assumptions in estimating expected losses:

- Used the Saffir-Simpson Scale, adjusted to 3 categories
- Hurricanes are independent of each other

Total expected loss: ***$4,231,960.57***
Estimated Loss Cost: ***12.87%***

Business Recommendations

1. Diversification of Business

- Underwriting guidelines
- Pricing Adjustments

2. Reinsurance

- Catastrophe Excess of Loss (CAT XL) Reinsurance
- Catastrophe Aggregate losses (Aggregate loss) Reinsurance

Diversification of Business

Pricing and Underwriting Adjustments

- Surcharge for CAT prone regions
- Premium discounts for customers who invest in resilience measures such as hurricane shutters, elevated structures, fire resistant materials

Engage in Reinsurance

1. **Catastrophe Excess of Loss (Cat XL) Reinsurance**

- Used for covering individual hurricanes
- Useful for isolating losses from hurricanes in the "high" category

2. **Catastrophe Aggregate losses (Aggregate loss) Reinsurance**

- Used for covering any hurricanes that happen throughout the year
- Useful for covering smaller but more frequent hurricanes

Appendix A: Expected Losses

- Utilized the Saffir Simpson Scale
- Researched historical hurricanes to estimate damage extent
- More resources required to create a probabilistic model

Saffir–Simpson scale

Category	Wind speeds (for 1-minute maximum sustained winds)			
	m/s	knots (kn)	mph	km/h
Five	≥ 70 m/s	≥ 137 kn	≥ 157 mph	≥ 252 km/h
Four	58–70 m/s	113–136 kn	130–156 mph	209–251 km/h
Three	50–58 m/s	96–112 kn	111–129 mph	178–208 km/h
Two	43–49 m/s	83–95 kn	96–110 mph	154–177 km/h
One	33–42 m/s	64–82 kn	74–95 mph	119–153 km/h

Appendix B: Policy Holder Reactions

- Risk-based pricing strategies
- Gradual rate adjustments
- Transparent communication

Appendix C: Average Hurricane Size

$$Area \approx \pi \cdot \frac{\Delta_{latitude} \cdot 69}{2} \cdot \frac{\Delta_{longitutde} \cdot 69cos(\mu_{longitude})}{2}$$

- Per longitude and latitude degree = 69 miles
- Division to convert to radius
- Cosine adjusts for Earth’s curvature as an ellipsoid